

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 39

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>Correct answer only.</i> • B (1)	1
01.2	<i>1 mark for each point.</i> • Number all possible sampling sites (eg on a grid map) (1) • Select the sites using a random number generator / table; (regular shape: grid coordinates from random numbers) (1)	2
01.3	<i>1 mark for each point.</i> • Sites may be chosen for convenience or because they appear to support (or dismiss) the hypothesis (1) • So the sample does not represent the whole area and conclusions are invalid (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	
02.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $38.0 / 0.002$ (1) • $= 19\,000$ (19 000 times / 1.9×10^4) (1)	2
02.2	<i>1 mark.</i> • Biomagnification (accept bioaccumulation as part of the process only if biomagnification is also implied by increase up the food chain) (1)	1

Question	Marking guidance	Marks
02.3	<i>1 mark for each point.</i> • Persistence: the pesticide is not broken down (by light, microbes, enzymes) so it remains in the environment / organisms long enough to pass along the food chain (1) • Liposolubility: the pesticide dissolves in lipids / fat so it is stored in body tissue and not excreted in (water-based) urine (1) • Each organism eats many organisms from the level below and retains the pesticide so concentration rises at each trophic level (1) • Herons have long lives so accumulate over time (1)	4
02.4	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Organochlorines killed non-target insects (bees, butterflies) and top predators (birds of prey, otters) through biomagnification (1) • They persist for decades so the damage continued long after use (1) • Use in houses (on walls / nets) does not enter food chains / no runoff into water, so does not biomagnify (1) • Benefit of preventing malaria deaths outweighs the risk (risk:benefit) (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-264, 266; AQA spec 3.4.3.2.8.	
03.1	<i>1 mark for each point.</i> • pH 5.6 (1) • Atmospheric carbon dioxide dissolves in rainwater to form (weak) carbonic acid (1)	2
03.2	<i>1 mark for each point.</i> • Limestone bedrock (calcium carbonate) is alkaline / basic and neutralises the acids as water percolates through it, so the lake pH stays high (7.2) (1) • Granite does not neutralise acids so lake water in A is acidic (4.6) (1) • Low pH mobilises / dissolves aluminium ions from the soil and rock (0.42 mg dm^{-3} in A) which are toxic to fish (damage gills) (1) • Acids also damage fish eggs / gills (exposed living cells) and dissolve the exoskeletons of invertebrate food; lichens and other food organisms lost (1)	4
03.3	<i>1 mark for each point.</i> • Corrodes metals: railway lines, railings, pipes, pylons, overhead power lines (1) • Dissolves / weakens limestone and marble buildings and statues (eg York Minster) (1)	2
03.4	<i>1 mark for each point.</i> • Sulfur dioxide is oxidised (by ozone / other oxidants) to sulfur trioxide (1) • Sulfur trioxide dissolves in water droplets to form sulfuric acid (SO₂ alone gives the weaker sulfurous acid) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 250-252; AQA spec 3.4.3.2.2.	

Question	Marking guidance	Marks
04.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • High-level: used fuel rods / fission products separated in reprocessing; vitrification in molten glass in sealed stainless steel containers, surrounded by concrete, air-cooled to remove decay heat (Sellafield) (2) • Intermediate: fuel cladding / metal tubes around fuel rods, filters from reprocessing; mixed with cement (encapsulated) in stainless steel drums, stored in concrete vaults (2) • Low-level solid: contaminated clothing, equipment; sealed in plastic bags inside steel drums / containers in a concrete-lined, monitored landfill (Drigg) (2)	6
04.2	<i>1 mark for each point.</i> • Radioactive decay of the concentrated fission products generates heat which could damage containers (1) • Isotopes with long half-lives (plutonium 24 400 years) remain hazardous for thousands of years (1)	2
04.3	<i>1 mark for each point.</i> • Liquids: filtered and passed through ion-exchange (sands / resins) to remove radionuclides, then discharged (monitored by CPA / CGM) (1) • Gases: filtered (HEPA / charcoal filters) to remove particles and iodine before release (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 303; AQA spec 3.4.3.2.14.	
05.1	<i>1 mark for each point.</i> • Molten iron from the blast furnace is converted directly to steel on the same site (1) • So it does not cool and need re-melting; transport energy between sites is also avoided; waste heat from one process used by another (eg Kalundborg) (1)	2
05.2	<i>1 mark for each point.</i> • Collecting and processing small dispersed quantities can use more energy in transport than is saved (1) • Mass production and large-scale mining have economies of scale; some materials (glass) take more energy to make than plastic but can be refilled many times (1)	2
05.3	<i>2 marks for each point (partial credit for a partly correct point).</i> • Transmitting at high voltage means low current so less energy is lost as heat in the cables (loss proportional to current squared); overhead cables cheaper than underground which need cooling (2) • ICT / smart grids co-ordinate real-time data on demand and supply from many sources, switching generation, storage (pumped storage, batteries) and flexible loads so surplus is not wasted and inefficient peaking plants are avoided (2)	4
05.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Proximity to the grid / demand to reduce transmission losses and cable costs (1) • Access to cooling water for thermal stations (1) • Resource availability (wind, tidal range, fuel deliveries) (1) • Land availability and planning / public acceptance; stable geology (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 234-235; AQA spec 3.3.4.2.3.	

Question	Marking guidance	Marks
06.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $(68\,000 - 7\,000) / 68\,000 \times 100$ (1) • = 89.7% (accept 90%) (1)	2
06.2	<i>1 mark for each point.</i> • Inflow from the rivers was reduced (sometimes to zero) so less water entered while evaporation continued (1) • Evaporation removes pure water leaving the dissolved salts behind, so they became concentrated in a smaller volume (1)	2
06.3	<i>1 mark for each point.</i> • Salinity rose above the range of tolerance of the freshwater fish species / osmotic stress; pollution by pesticides, fertilisers and industrial waste concentrated in the lake (1) • Exposed lake bed of salt and pesticide-contaminated dust blown by wind onto farmland and settlements (health problems, soil salinisation) (1) • Loss of wetland habitats and wildlife; local climate more extreme (loss of the moderating effect of the water body) (1) • Loss of livelihoods; conflict between the countries sharing the rivers (1)	4
06.4	<i>1 mark for each point.</i> • Upstream abstraction for irrigation reduces the flow available downstream (1) • Upstream HEP dams hold water in summer for winter generation, reducing summer irrigation water downstream; growing populations increase demand (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 117-118; AQA spec 3.2.2.1-3.2.2.2.	
07.1	<i>1 mark for each point.</i> • 2020 has higher peak flow in the wet season (32 vs $22\text{ m}^3\text{ s}^{-1}$ in March) (1) • 2020 has much lower dry-season flow (2 vs $9\text{ m}^3\text{ s}^{-1}$ in July / August) (1) • Flow is much more variable / seasonal in 2020; the river almost dries up (1)	3
07.2	<i>1 mark for each point.</i> • In 1990 snow / ice stored wet-season precipitation and released it gradually as meltwater during the dry season, evening out the flow (1) • With less ice, precipitation now falls as rain / melts immediately and runs off quickly, increasing wet-season flow and flood risk (1) • Little stored ice remains to melt in the dry season so flow falls very low (1)	3
07.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Water shortages for irrigation / drinking / HEP in the dry season (1) • Flooding in the wet season (1) • Sudden floods from ice-lake outbursts (glacial lake outburst floods) (1)	2
07.4	<i>1 mark for each point.</i> • Surface meltwater flows down cracks to the base, lubricating the ice so it slides faster; if this exceeds the faster melting at the front, the front advances (1) • But if snowfall at the source has not increased, total volume falls; both retreating and extending glaciers can indicate warming (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 94-96; AQA spec 3.2.1.2.	

Question	Marking guidance	Marks
08.1	<i>1 mark.</i> • As temperature increases the maximum dissolved oxygen concentration decreases (non-linear: falls faster at low temperatures) (1)	1
08.2	<i>1 mark for the correct answer.</i> • About 39 °C (accept 38-40) (1)	1
08.3	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • 9.1 - 7.9 (value at 27 °C, accept 7.8-8.0) (1) • = 1.2 mg dm ⁻³ (accept 1.1-1.3) (1)	2
08.4	<i>1 mark for each point.</i> • Figure 1 shows the maximum / saturation values; processes that remove oxygen (aerobic decomposition of organic matter / respiration) can give values below the line (1) • Higher temperature increases the metabolic rate of organisms and the rate of bacterial decomposition so oxygen is used faster; values above the line are impossible (1)	2
08.5	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Increased metabolic rate so more energy used / less surplus for growth (1) • Eggs develop faster / hatch earlier (out of step with food supply) (1) • Non-indigenous warm-water species may thrive and out-compete indigenous species (1) • Toxicity of other pollutants is greater as toxins are metabolised faster (1) • Reduced resistance to disease (1) • Proteins / enzymes denatured if temperature is very high (above about 40 °C) (1)	3
08.6	<i>1 mark.</i> • Hot water is sprayed from a shower floor in a tall hollow tower; air absorbs the heat and rises, drawing in more air (updraught), so the water returned is cooler (and saturated with oxygen) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 256-257; AQA spec 3.4.3.2.6.	
09.1	<i>1 mark for each point.</i> • 1 nitrogen fixation; 2 nitrification (1) • 3 denitrification; 4 ammonification (1) • 5 ionisation; 6 leaching (3 marks for all six, 2 for four or five, 1 for two or three) (1)	3
09.2	<i>1 mark for each point.</i> • Drainage makes the soil aerobic (1) • Aerobic nitrifying bacteria (and nitrogen fixers) increase so more ammonium is converted to nitrate (1) • Anaerobic denitrifying bacteria decline so less nitrate is lost as nitrogen gas (1)	3
09.3	<i>1 mark for each point.</i> • Nitrogen gas is very unreactive so most organisms cannot use it directly (1) • Only nitrogen fixation (bacteria, lightning, Haber process) converts it to usable ammonium / nitrate; the cycle recycles a small usable amount (1)	2

Question	Marking guidance	Marks
09.4	<i>1 mark for each point.</i> • Rhizobium bacteria in their root nodules fix atmospheric nitrogen into ammonium / proteins (1) • When the plants die / are ploughed in, decomposition and nitrification release nitrates for the following crop (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 152-154; AQA spec 3.2.4.3.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Context: Lasky's principle - vast quantities exist at low grade; reserves vs resource; COOG affected by price, technology, energy costs • Improved exploration: higher resolution multispectral satellite sensors, IR spectroscopy, gravimetry, magnetometry, seismic, resistivity, portable field analysis, targeted trial drilling • Mechanisation allowing deeper underground mines and larger open-cast mines • Low-energy extraction of low-grade ores: bioleaching with acidophilic bacteria and fungi, phytomining, iron displacement, bacterial adsorption of rare earths, polymer adsorption from seawater • Previously inaccessible deposits: polymetallic nodules, deep sea sediments, Antarctica (currently protected) - environmental and legal issues • Extending use: recycling (advantages, difficulties), Cradle to Cradle design, product lifetime extension, substitution of scarce materials, circular economy • Evaluation: environmental impacts vs benefits; energy costs; economics; which strategies are most sustainable; conclusion	9
	Model answer - Level 3 (8 marks)	
	Lasky's principle means that as grade falls, the quantity of metal available rises enormously, so depletion of high-grade deposits does not mean running out - it means the cut-off ore grade must fall, which depends on price, technology and energy costs. Better exploration finds what remains: high-resolution multispectral and infrared satellite sensors, gravimetry, magnetometry, seismic and resistivity surveys and portable field analysers narrow the search so expensive trial drilling is targeted. Mechanisation allows deeper underground mines and larger open-cast pits, and previously inaccessible deposits - polymetallic nodules and deep-sea sediments, and Antarctica if its protection were ever lifted - become possible, with serious environmental and legal problems.	

Question	Marking guidance	Marks
	The most sustainable route to low-grade ores is low-energy extraction: bioleaching with acidophilic bacteria and fungi, phytomining, iron displacement of copper, bacterial adsorption of rare earths and polymer adsorption from seawater all avoid the energy of crushing and smelting vast tonnages. Equally important is extending the use of metal already mined: recycling, Cradle-to-Cradle design so products can be disassembled, longer product lifetimes and substitution of scarce metals - the circular economy approach.	
	Each strategy has costs: deep-sea mining and Antarctica risk unique habitats, deeper mines need more energy, recycling is limited by dispersal and mixed alloys. The strategies that mimic natural low-temperature processes and keep metals in use are the most sustainable, and exploration and new mining should be the last resort rather than the first.	
	Weaker answer - Level 1 (3 marks)	
	When high grade ores run out we can mine lower grade ores instead, but this uses more energy. Better technology like satellites can find new deposits. We could mine the deep sea or Antarctica. Bacteria can be used to extract metals from low grade ore. Recycling metals means we do not need to mine as much. These strategies will help secure supplies of metals in the future.	
	What makes the difference	
	• Level 1 names most strategy types but Level 3 explains each (what the surveys measure, why low-temperature extraction matters, what Lasky's principle implies).	
	• Level 3 weighs environmental and energy costs and ranks the strategies; Level 1 offers no evaluation.	
	• Framing the answer with the concept of cut-off ore grade shows understanding rather than recall.	
10.2	<i>1 mark.</i> • All of the material that exists in the lithosphere, including that which will never be technologically or economically exploitable (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 130-147; AQA spec 3.2.3.	

Question	Marking guidance	Marks
11.1	Discuss the strategies that can be used to provide sustainable supplies of water for a growing population. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Context: uneven distribution of water, growing population and per capita demand, unsustainable exploitation of rivers and aquifers, hydrological cycle links • New sources: rainwater collection, rivers (discharge, fluctuations, contaminants), reservoirs and site selection, estuary barrages, desalination, unexploited aquifers • Management: artificial aquifer recharge, river-regulation reservoirs, inter-basin transfers, afforestation • Conservation: low-water appliances, xeriscaping, drip irrigation, grey water, leak reduction, metering; pollution control to protect sources • Treatment: matching quality to use; screens, sedimentation, aeration, flocculation, filtration, sterilisation, pH adjustment, activated carbon, reverse osmosis • Evaluation of costs, energy, environmental impacts (habitat loss, methane, inter-tidal habitats, conflicts) and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-127; AQA spec 3.2.2.4-3.2.2.5.	25

Question	Marking guidance	Marks
11.2	Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Historical data limitations; proxy data - ice cores, isotope ratios (O-18:O-16, C-14), Mg:Ca in calcite, tree rings, pollen, coral • Satellite sensors: visible light imagery, infrared for vegetation and temperature, radar microwaves for wind, wave height, sea level and oil, weather monitoring; passing through cloud • Sensor carriers: ROVs / drones, AUVs, balloons, aircraft, animals; data recorders for abiotic factors; GPS tracking of floats and sampling sites; Argo floats and ocean monitoring • Climate models and their assumptions; monitoring atmospheric composition (Mauna Loa), ozone (Dobson units), sea level and ice • Applications to hydrosphere, minerals (remote sensing surveys), soils, pollution monitoring • Evaluation: improved coverage and resolution vs cost, calibration, interpretation, remaining uncertainty; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 9, 89-107, 402-407; AQA spec 3.2, 3.7.	25